Week 14: Intentionality, the Intentional Stance and Can Machines Think?

Philosophy of Artificial Intelligence

Philosophy of Artificial Intelligence (11118BLG003). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Mental states are about things: Beliefs are about the world and desires are for outcomes. This week asks whether machines can have states with this aboutness and examines Dennett's intentional stance and its formalisation as inverse planning. It closes the course by returning to the question of whether machines can think [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to explain intentionality and the distinction between original and derived intentionality. They should be able to describe the physical, design and intentional stances and to compute a goal posterior by inverse planning. Finally, they should be able to defend a reasoned position on whether machines can think, drawing on the whole course.

Lecture

The mark of the mental

Brentano proposed that what distinguishes mental phenomena is their intentionality, their being directed at an object [1]. A belief is about something, a desire is for something and a fear is of something, even when the object does not exist. Searle distinguished intrinsic, or original, intentionality from derived intentionality [4]. The words on a page and the states of a computer have derived intentionality: They are about things only because people interpret them. Minds have original intentionality, which does not depend on interpretation. On Searle's view, the symbols of a program have only derived intentionality, which connects this week with the Chinese Room of Week 4.

The intentional stance

Dennett proposed a different approach [2]. There are three stances from which the behaviour of a system can be predicted. The physical stance uses physics and the system's physical constitution. The design stance assumes that the system will behave as it was designed to behave. The intentional stance treats the system as a rational agent with beliefs and desires and predicts that it will act to further its goals in the light of its beliefs. A system is an intentional system if its behaviour is reliably and voluminously predictable from the intentional stance. Dennett argued that beliefs are real in the way that patterns are real: They capture regularities in behaviour that cannot be captured more economically from lower-level stances [5]. The view sits between the realism of Searle and the view that talk of beliefs is merely useful fiction.

Attributing minds

People attribute intentions readily. Heider and Simmel showed participants a short film of geometric shapes moving around a box [6]. Almost all of them described the film in terms of intentions and emotions, such as a large triangle bullying a small one. Baker, Saxe and Tenenbaum formalised such attributions as inverse planning [3]. An observer assumes that an agent acts approximately rationally towards a goal and uses Bayesian inference to work backwards from observed actions to the most probable goal. The model predicted human judgements of goals closely in their experiments. Figure 14.1 shows how the posterior over three goals sharpens as more of a path is observed. The interactive lab uses this model and compares the predictions of the intentional stance with those of a simple physical stance.

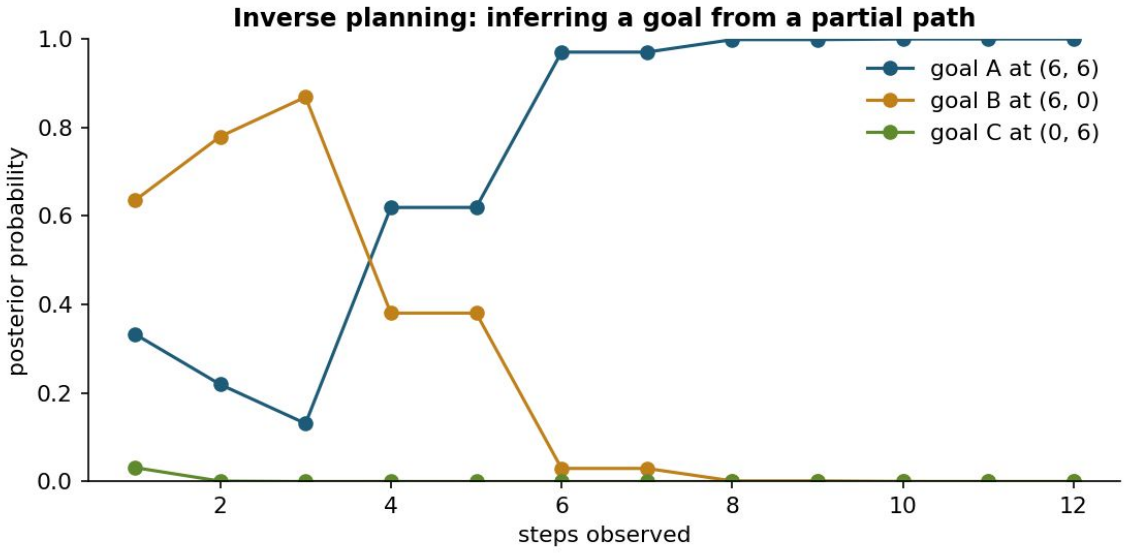


Figure 14.1. Posterior probabilities of three possible goals as successive steps of a path are observed, computed by Bayesian inverse planning [3].

Can machines think? The course in review

The intentional stance offers one answer to the question with which the course began. If adopting the stance towards a system yields reliable predictions, the system has beliefs and

desires in the only sense that Dennett thinks matters. Many present AI systems qualify for a limited range of behaviour. Searle would reply that predictive usefulness does not create original intentionality, just as the Chinese Room produced appropriate behaviour without understanding. Shanahan's caution applies here too: The intentional stance towards a dialogue system may predict well while misleading about what the underlying model is [7].

The course has presented several ways to sharpen the question. Behavioural tests, such as Turing's, can be passed without settling it. Arguments about computation, grounding, systematicity, embodiment, relevance and limits bear on what thinking requires. Theories of consciousness address whether there is anything it is like to be the system. Studies of understanding, creativity and intentionality show where present systems succeed and where claims outrun evidence. The final essay asks each student to take a position and defend it with these tools.

> ***Pause and reflect.*** *After fourteen weeks, has your answer to the question whether machines can think changed? Which week changed it most?*

Hands-on components

Interactive lab: Part A animates two shapes moved by very simple rules; describe what happens before revealing the rules. Part B lets you move an agent in a grid and computes the posterior over three goals, comparing the next-move predictions of the physical and intentional stances [2, 3, 6]. Link: https://utkukose.github.io/philosophy-of-ai-NB-lecture/weeks/week-14/lab.html

Colab notebook: It implements Bayesian inverse planning in a grid, simulates goal-directed agents and compares how well the physical and intentional stances predict their next moves [2, 3]. Link: https://colab.research.google.com/github/utkukose/philosophy-of-ai-NB-lecture/blob/main/weeks/week-14/NB14_intentional_stance.ipynb

Reflection prompts

1. Describe what you saw in Part A before revealing the rules. What does your description reveal about the intentional stance?
2. Do present AI assistants satisfy Dennett's criterion for intentional systems? For which behaviours, and where does the stance fail?
3. State and defend your final answer to the question whether machines can think, citing at least three weeks of the course.

Weekly task and submission

What to do. Write a short position essay on the question of the course: Can machines think? First, apply the intentional stance to one AI system and say how well it predicts the system's

behaviour. Second, use Searle's distinction between original and derived intentionality to ask whether the states of the system are about anything for the system itself. Third, take a clear position and support it with arguments from at least three earlier weeks. Finally, answer the strongest objection to your position.

Where to look. Lecture: sections "The mark of the mental", "The intentional stance" and "Can machines think? The course in review". Interactive lab: Part A, where moving shapes invite descriptions in terms of intentions, and Part B on inverse planning. Colab notebook: Section 2, which compares how well the physical and intentional stances predict the next move.

Example. The intentional stance predicts a navigation app well: It wants to reach the destination and believes that a road is closed. Searle would reply that this intentionality is derived from the designers and users of the app. A student could accept Dennett's view for prediction and still side with Searle on whether the app thinks. The Chinese Room of Week 4 and the grounding problem of Week 5 could support this position.

What to submit. About 600 words that cite all sources in square brackets, and the Colab notebook with Exercises 1 and 2 completed. This weekly task is separate from the final capstone, whose page lists its own deliverables.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

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[5] Dennett, D. C. (1991). Real patterns. The Journal of Philosophy, 88(1), 27-51. https://doi.org/10.2307/2027085

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